报告题目与摘要

陈张弛 （华东师范大学）
题目： Hypercyclic holomorphic mappings with slow growth
摘要： An entire function h: C to C is called hypercyclic if for some non-zero complex number a, the tranlation orbit {h(.+n*a)} is dense in the space of entire functions with compact-open topology. Such hypercyclicity can also be stated for holomorphic maps h: X to Y with respect to some generalized translations. Dinh-Sibony posed a question to determine the slowest Nevanlinna growth of hypercyclic entire functions C to C or entire curves C to P^1. We answered this question by constructing hypercyclic entire curves into P^n with the optimal slow growth rate. Furthermore, we construct frequently hypercyclic entire curves into P^n with the optimal slow growth rate. These are joint works with Bin Guo, Dinh Tuan Huynh and Song-Yan Xie.

丁聪（深圳大学）
题目：Rigidity of Schubert classes and its application in holomorphic isometry
摘要：Schubert classes form the additive basis for the homology group of a flag variety. A classical problem in algebraic geometry is to determine the representatives of these Schubert classes. In this talk, we will give several notions of rigidity associated with this problem, along with our recent results. Furthermore, we will discuss the applications of some rigidity results to the study of holomorphic isometries in several complex variables. Part of this talk is based on a joint work with Qifeng Li.

蒋天枢（中国科学技术大学）
题目：TBA
摘要：TBA

韩骥原（西湖大学）
题目：On the existence of weighted-cscK metrics
摘要：Weighted-cscK metrics provide a universal framework for the study of canonical metrics, e.g., extremal metrics, Kahler-Ricci soliton metrics, \mu-cscK metrics. In joint works with Yaxiong Liu, we prove that on a Kahler manifold X, the G-coercivity of weighted Mabuchi functional implies the existence of weighted-cscK metrics. In particular, there exists a weighted-cscK metric if X is a projective manifold that is weighted K-stable for models. We will also discuss some progress on singular varieties.

胡佳俊（清华大学）
题目：An algebro-geometric approach to the extremals of log-concave inequalities
摘要：Log-concavity is a ubiquitous phenomenon, yielding fundamental inequalities across diverse areas of mathematics. This naturally leads the problem of characterizing the equality cases. Our focus is the counterpart of this problem in algebraic geometry: the characterization of the extremals of the Khovanskii-Teissier inequality. We provide a complete solution to this problem for semiample divisors. As a key application, this allows us to solve the corresponding

problem in convex geometry, providing the characterization of the extremals of the Alexandrov-Fenchel inequalities for rational polytopes. This is joint work with Jian Xiao.

黄鹏飞（南京大学）
题目：Symmetric spaces for groups over involutive algebras and applications
摘要：Symplectic groups and orthogonal groups over noncommutative involutive algebras provide new way of realizing classical Lie groups. When in particular, the involutive algebra is Hermitian or the complexification of a Hermitian algebra, one can identify the corresponding maximal compact subgroups. This new perspective allows for the realization of various geometric models for the symmetric space of these groups. In this talk, we will try to illustrate such ideas and describe the tangent spaces for each of the models, as well as the application to Higgs bundles. This is a joint work with Kydonakis, Rogozinnikov and Wienhard.

李超（南京理工大学）
题目：Gromov-Hausdorff limits and Holomorphic isometries
摘要：Gromov’s precompactness theorem for Riemannian manifolds and its refined versions have a profound impact on differential geometry. In this talk, I will introduce a recent joint work with Claudio Arezzo and Andrea Loi. We established a compactness theorem for complete immersed Kahler submanifolds in a fixed ambient space, and then applied it to study Kahler submanifolds of projective spaces.

欧文浩（中国科学院）
题目：凯勒双有理几何
摘要：我们将介绍最近关于紧凯勒流形上双有理几何的进展，包括有理曲线存在性等问题。这是基于近期和 Das，傅鑫的合作。

沈洋（重庆理工大学）[PPT，12/26-30]
题目：Penrose transformations of automorphic cohomology groups on flag domains
摘要：In this talk, I will present recent joint work with Kefeng Liu on the Penrose transformation of automorphic cohomology groups of homogeneous line bundles on flag domains. We construct the Penrose transforms explicitly and identify the geometric and representation-theoretic conditions under which the Penrose transforms are isomorphisms. As an application, we show that higher automorphic cohomology groups of certain line bundles on non-classical flag domain are canonically isomorphic to spaces of automorphic forms on Hermitian symmetric domains. We also propose the problem concerning the arithmetic structure of totally degenerate limits of discrete series (TDLDS) on non-classical flag domains, pointing to further connections with geometric representation theory and Hodge theory.

汤凯（浙江师范大学）
题目：On Hermitian manifolds with constant curvature
摘要：A long-standing conjecture predicts that a compact Hermitian manifold with

constant holomorphic sectional curvature λ is Kaehler if $\lambda \neq 0$ and Chern-flat if $\lambda = 0$. We will discuss the validity of this conjecture under additional conditions. Similarly, we study the constant mixed curvature $C(a,b)=\lambda$, which is a convex combination of Ricci curvature and holomorphic sectional curvature introduced by Chu-Lee-Tam. We prove that if a compact Hermitian surface with constant mixed curvature λ, then the Hermitian metric must be Kaehler unless $\lambda = 0$ and $2a+b=0$. We also consider general kth-mixed curvature for Hermitian manifolds.

王枫（浙江大学）
题目：Stability of Fano spherical varieties
摘要：We define a modified Futaki invariant and give an expression in terms of intersection numbers. We will show the equivalence of different notions of stability and give a stability criterion on Q-Fano spherical varieties.

吴菊杰(中山大学-珠海)
题目：Equivalence between VMO functions and Zero Lelong number functions
摘要：We prove that a plurisubharmonic function on a domain in the complex Euclidean space is a locally VMO (Vanishing Mean Oscillation) function if and only if its Lelong number at each point vanishes. We also give a global version of this result when the boundary of the domain satisfies the \textit{interior sphere condition}. An example emphasizes the importance of this condition. These equivalences contribute to a better understanding of the behavior of singular plurisubharmonic functions. We end the paper by discussing the link between the residual Monge-Amp\`ere mass and VMO functions, by providing examples. This is joint work with Severine Biard.

徐旺（中山大学）
题目：On the Converse of Prekopa's Theorem and Berndtsson's Theorem
摘要：Extending the framework of the converse L^2 theory, we establish converse results for Berndtsson's theorem on the plurisubharmonic variation of Bergman kernels, showing that both the plurisubharmonicity of functions and the pseudoconvexity of domains are necessary conditions in twisted senses. We also prove analogous results for Berndtsson's theorem on the positivity of direct images and Prekopa's theorem from convex analysis. This is a joint work with Dr. Hui Yang.

张德凯（华东师范大学）
题目：The quaternionic form type Calabi-Yau equation on a compact hyperKahler manifold
摘要：The form type Calabi-Yau equation was introduced by Fu-Wang-Wu 2010 to find balanced metrics. It is related to the Gauduchon conjecture which has been solved by Szekelyhidi-Tosatti-Weinkove. In this talk, we consider the quaternionic form type Calabi-Yau equation on compact Hermitian manifolds. We prove the existence of the smooth solution on a compact hyperKahler manifold. This is a joint work with Prof. Jixiang Fu and Dr. Xin Xu.

张润泽(武汉大学 & Université Côte d'Azur)
题目：Unobstructed deformations for pairs: a log ddbar lemma
摘要： The ddbar-lemma is a powerful tool in complex and algebraic geometry. In the first part of my talk, I will establish a logarithmic ddbar-type lemma on compact Kähler manifolds for logarithmic differential forms with values in the dual of a pseudo-effective line bundle. In the second part, I will present an application to the work of L. Katzarkov－M. Kontsevich－T. Pantev on unobstructed locally trivial deformations of generalized log Calabi－Yau pairs with weights, extending their projective result to the broader Kähler setting.